

```
import pandas as pd

# Load the dataset
df = pd.read_csv('/content/nba-players.csv.csv')

# Display the first 5 rows of the DataFrame
display(df.head())
```

	Unnamed: 0	name	gp	min	pts	fgm	fga	fg	3p_made	3pa	...	fta	ft
0	0	Brandon Ingram	36	27.4	7.4	2.6	7.6	34.7	0.5	2.1	...	2.3	69.9
1	1	Andrew Harrison	35	26.9	7.2	2.0	6.7	29.6	0.7	2.8	...	3.4	76.5
2	2	JaKarr Sampson	74	15.3	5.2	2.0	4.7	42.2	0.4	1.7	...	1.3	67.0
3	3	Malik Sealy	58	11.6	5.7	2.3	5.5	42.6	0.1	0.5	...	1.3	68.9
4	4	Matt Geiger	48	11.5	4.5	1.6	3.0	52.4	0.0	0.1	...	1.9	67.4

5 rows × 22 columns

✓ Define the Target Variable

The `target_5yrs` column indicates whether a player lasted more than 5 years in the NBA. This will be our dependent variable for prediction.

```
# Display the distribution of the target variable
display(df['target_5yrs'].value_counts())
```

	count
target_5yrs	
1	831
0	509

dtype: int64

✓ Drop Non-Predictive Columns

Columns such as player names ('name') and an arbitrary index ('Unnamed: 0') are non-predictive and can lead to data leakage or add noise to the model. We will drop these columns to prepare our dataset for feature engineering.

```
# Drop non-predictive columns
df = df.drop(columns=['Unnamed: 0', 'name'])

# Display the first few rows of the cleaned DataFrame
display(df.head())
```

	gp	min	pts	fgm	fga	fg	3p_made	3pa	3p	ftm	fta	ft	oreb	dreb	reb
0	36	27.4	7.4	2.6	7.6	34.7	0.5	2.1	25.0	1.6	2.3	69.9	0.7	3.4	4.1
1	35	26.9	7.2	2.0	6.7	29.6	0.7	2.8	23.5	2.6	3.4	76.5	0.5	2.0	2.5
2	74	15.3	5.2	2.0	4.7	42.2	0.4	1.7	24.4	0.9	1.3	67.0	0.5	1.7	2.2
3	58	11.6	5.7	2.3	5.5	42.6	0.1	0.5	22.6	0.9	1.3	68.9	1.0	0.9	1.9
4	48	11.5	4.5	1.6	3.0	52.4	0.0	0.1	0.0	1.3	1.9	67.4	1.0	1.5	2.5

✓ Correlation Analysis

We will perform a correlation analysis to identify features that are highly correlated with each other (multicollinearity) and with the target variable (`target_5yrs`). This helps in understanding feature relationships and potentially reducing redundancy.

```
import seaborn as sns
import matplotlib.pyplot as plt

# Calculate the correlation matrix
corr_matrix = df.corr()

# Display correlations with 'target_5yrs'
display(corr_matrix['target_5yrs'].sort_values(ascending=False))

# Plot a heatmap of the correlation matrix for a visual overview
plt.figure(figsize=(15, 12))
sns.heatmap(corr_matrix, annot=True, cmap='coolwarm', fmt=".2f")
plt.title('Correlation Matrix of NBA Player Statistics')
plt.show()
```


	target_5yrs
target_5yrs	1.000000
gp	0.396833
min	0.317805
fgm	0.317594
pts	0.315981
reb	0.299406
ftm	0.296841
fta	0.296089
oreb	0.293307
fga	0.292660
dreb	0.284677
tov	0.272348
stl	0.229811
fg	0.227134
blk	0.210114
ast	0.175353

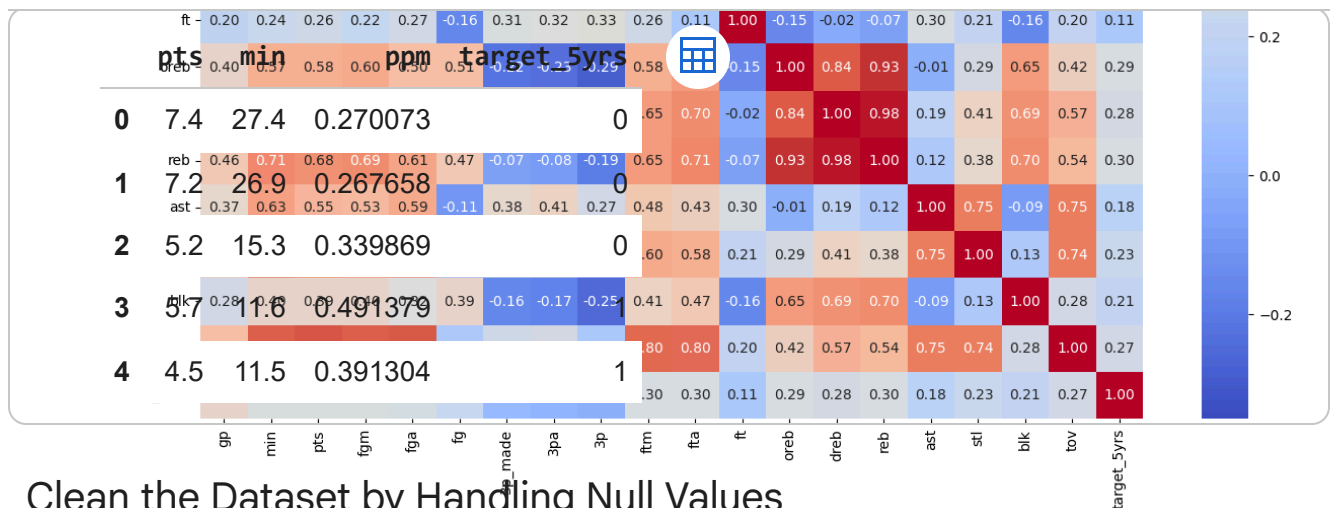
▼ Engineer a New Composite Feature: Points Per Minute (PPM)

To create a more insightful metric, we will engineer 'Points Per Minute' (PPM) by dividing a player's total points ('pts') by their total minutes played ('min'). This normalizes scoring by playing time, offering a per-minute efficiency measure.

```
# Engineer 'Points Per Minute' (PPM)
# Handle potential division by zero if 'min' is 0, by setting PPM to 0 in those
df['ppm'] = df.apply(lambda row: row['pts'] / row['min'] if row['min'] > 0 else
```

```
# Display the first few rows with the new feature
display(df[['pts', 'min', 'ppm', 'target_5yrs']].head())
```





✓ Clean the Dataset by Handling Null Values

Before building a machine learning model, it's essential to handle missing values. We will identify columns with null values and decide on an appropriate strategy (e.g., imputation, removal) to ensure the dataset's integrity and readiness for modeling.

```
# Identify columns with null values and count them
missing_values = df.isnull().sum()
missing_values = missing_values[missing_values > 0].sort_values(ascending=False)

display("Columns with missing values:")
display(missing_values)
```

```
'Columns with missing values:'
```

```
0
```

```
dtype: int64
```

Documentation of Feature Engineering Choices

To summarize the feature engineering pipeline so far, we have:

1. **Defined Target Variable:** Clearly identified `target_5yrs` as our dependent variable.
2. **Dropped Non-Predictive Columns:** Removed 'Unnamed: 0' and 'name' to prevent data leakage and reduce noise.
3. **Performed Correlation Analysis:** Examined correlations between features and with the target variable to understand relationships and identify potential multicollinearity.
4. **Engineered Composite Feature:** Created 'Points Per Minute' (`ppm`) to normalize player scoring by playing time, offering a more efficient metric.